

Lecture/Week - 7

Numerical Differentiation

Our focus will be on

→ 1st Order Derivative.
2nd " " " "

First Derivative

Note :- Derivative of a function $f(x)$ is another function $f'(x)$.

Example $f(x) = x^3 + 4x^2 - 1$

$$\boxed{f'(x) = 3x^2 + 8x}$$

or $f(x) = \sin x + x^2$

$$f'(x) = \cos x + 2x$$

Can we approximate/find a function numerically?

Numerically we can only approximate a number ~~number~~.

This means we can find $f'(a)$

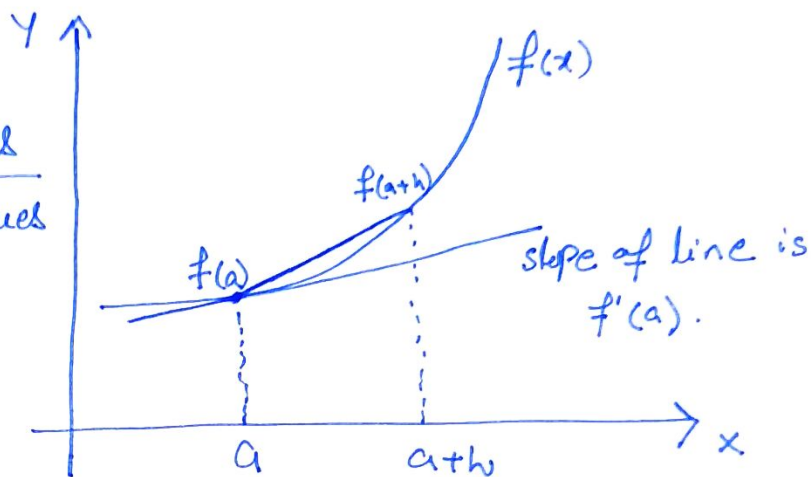
Derivative at a point.

(2)

First derivative

The aim is to approximate the slope of a curve ' $f(x)$ ' at a particular point ' $x=a$ '.

$m = f'(a) \approx \frac{\text{difference in } y\text{-values}}{\text{difference in } x\text{-values}}$
 slope



$$f'(a) \approx \frac{f(a+h) - f(a)}{a+h - a}$$

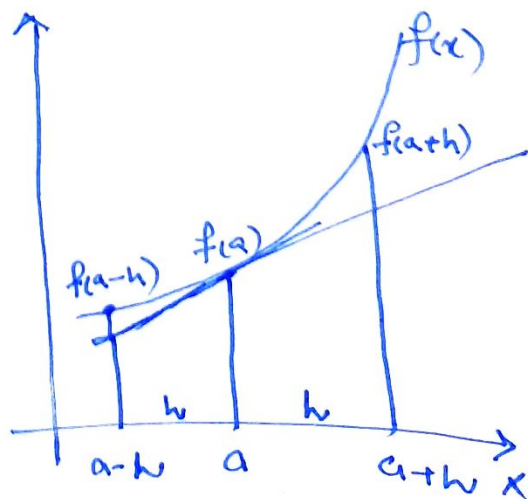
$$f'(a) \approx \frac{f(a+h) - f(a)}{h} \quad \text{--- (1)}$$

↳ Forward difference approximation to the derivative of f at $x=a$.

$$m = f'(a) \approx \frac{f(a) - f(a-h)}{a - (a-h)}$$

$$f'(a) \approx \frac{f(a) - f(a-h)}{h} \quad \text{--- (2)}$$

↳ Backward difference approximation of $f'(a)$



Taking average of (1) and (2)

$$f'(a) \approx \frac{\frac{f(a+h) - f(a)}{h} + \frac{f(a) - f(a-h)}{h}}{2}$$
$$= \frac{f(a+h) - \cancel{f(a)} + \cancel{f(a)} - f(a-h)}{2h}$$

$$\boxed{f'(a) = \frac{f(a+h) - f(a-h)}{2h}} \rightarrow (3)$$

↳ Central difference approximation of $f'(a)$.

(3)

(4)

Example ①Approximate derivative of $\cos x$ at $x = \pi/3$

a) $h = 0.1$ b) $h = 0.01$ c) $h = 0.001$ d) $h = 0.0001$

Use forward difference method.

Solution

$$f(x) = \cos x \quad a = \pi/3$$

a) $h = 0.1 \quad f'(a) = \frac{f(a+h) - f(a)}{h} = \frac{\cos(a+h) - \cos(a)}{h}$

$$= \frac{\cos(\pi/3 + 0.1) - \cos(\pi/3)}{0.1} = \frac{0.41104381 - 0.5}{0.1} = \underline{\underline{-0.88956192}}$$

b) $h = 0.01 \quad f'(a) = \frac{f(a+h) - f(a)}{h} = \frac{\cos(a+h) - \cos a}{h}$

$$= \frac{\cos(\pi/3 + 0.01) - \cos(\pi/3)}{0.01} = \frac{0.49131489 - 0.5}{0.01} = \underline{\underline{-0.86851095}}$$

c) $h = 0.001$

$$\frac{\cos(\pi/3 + 0.001) - \cos(\pi/3)}{0.001} = \frac{0.49913372 - 0.5}{0.001} = \underline{\underline{-0.86627526}}$$

d) $h = 0.0001$

$$= \underline{\underline{-0.86605040}}$$

Exact Solution

$$f'(x) = -\sin x$$

$$f'(\pi/3) = -\sin(\pi/3) = \underline{\underline{-0.86602540}}$$

Smaller the value of 'h' better will be our approximations.

Example ② Approximate derivative of $f(x) = \cos x$ at $x = \pi/3$ using central difference formula for

- a) $h=0.1$ b) $h=0.01$ c) $h=0.001$ d) $h=0.0001$

Solution

$$f(x) = \cos x$$

$$a = \pi/3$$

a) $h=0.1$

$$\begin{aligned} f'(a) &= \frac{f(a+h) - f(a-h)}{2h} = \frac{\cos(a+h) - \cos(a-h)}{2h} \\ &= \frac{\cos(\pi/3 + 0.1) - \cos(\pi/3 - 0.1)}{2(0.1)} = \frac{0.41104381 - 0.58396036}{2(0.1)} \\ &= \underline{\underline{-0.86458275}} \end{aligned}$$

b) $h=0.01$

$$\begin{aligned} f'(a) &= \frac{f(a+h) - f(a-h)}{2h} = \frac{\cos(\pi/3 + 0.01) - \cos(\pi/3 - 0.01)}{2(0.01)} \\ &= \frac{0.49131489 - 0.50863511}{0.02} = -0.86601097 \end{aligned}$$

c) $h=0.001$

$$= -0.86602526$$

d) $h=0.0001$

$$= -0.86602540$$

Central Difference Method give better approximation when compared with forward or backward difference method.

Second Derivative

$$f''(a) = \frac{f(a+h) - 2f(a) + f(a-h)}{h^2}$$

Central difference formula for 2nd derivative

Example

The distance 's' of runner from a fixed point is measured at intervals of half a second.

t	0	0.5	1.0	1.5	2.0
s	0	3.65	6.80	9.90	12.15

Obtain runner's velocity at $t = 0.5$ sec, $t = 1.25$ sec, $t = 0.7$ sec
 $t = 2.0$ sec.

Solution

$$v(0.5) = s'(0.5) = ?$$

$$v(1.25) = s'(1.25) = ?$$

$$v(0.7) = s'(0.7) = ?$$

$$v(2.0) = s'(2.0) = ?$$

$$v(0.5) = s'(0.5) = \frac{s(a+h) - s(a-h)}{2h}$$

$$h = 0.5$$

$$a = 0.5$$

$$= \frac{s(0.5+0.5) - s(0.5-0.5)}{2(0.5)} = \frac{s(1.0) - s(0.0)}{2(0.5)}$$

$$= \frac{6.80 - 0}{1} = 6.80 \text{ m/sec.}$$

$$V(1.25) = f'(1.25) = \frac{f(a+h) - f(a-h)}{2h} \quad \begin{matrix} a = 1.25 \\ h = 0.25 \end{matrix}$$

$$= \frac{f(1.25 + 0.25) - f(1.25 - 0.25)}{2(0.25)}$$

$$= \frac{f(1.5) - f(1.0)}{2(0.25)} = \frac{9.90 - 6.80}{2(0.25)} = 6.20 \text{ m/sec.}$$

$V(0.7) = f'(0.7)$ Not possible with this method
Have to use Interpolation

→ $f(0.7)$ not given

→ h is not same.

$V(2.0) = f'(2.0)$ cannot use central and forward difference method.

Backward difference.

$$f'(2.0) = \frac{f(2.0) - f(2.0 - 0.5)}{0.5} \quad h = 0.5$$

$$= \frac{12.15 - 9.90}{0.5} = \text{---} \text{Ans'}$$

(8)

Approximate runner's acceleration at $t = 1.5$

$$a(1.5) = s''(1.5) = \frac{s(a+h) - 2s(a) + s(a-h)}{h^2}$$

$$\begin{aligned} a &= 1.5 \\ h &= 0.5 \end{aligned}$$

$$= \frac{s(1.5+0.5) - 2s(1.5) + s(1.5-0.5)}{(0.5)^2}$$

$$= \frac{s(2.0) - 2s(1.5) + s(1.0)}{(0.5)^2} = \frac{12.15 - 2(9.90) + 6.80}{(0.5)^2}$$

$$= -3.40 \text{ m/sec}^2$$

Ans.